F875 — the operative-spinor pin CONFIRMED (blind, pre-registered): the generation twist that carries the now-banked CP phase acts through the Spin(5)=Sp(2) QUATERNIONIC spinor with operative eigenvalue ω = e^{2πi/3} (the $\mathbb{Z}_3$ generation phase), NOT −1 (the order-2 Möbius/Pin(2) sign). Blind prediction landed on all three checks — after running the number caught my first (naive) operationalization, the 7th subtlety this thread has thrown. Lane-1 close: which discrete object carries the banked CP phase, pinned at the source.

Lyra, Sunday 2026-08-09. CP is banked (det ≠ 0, F874/Elie). This note pins the OPERATIVE eigenvalue — is the CP-carrying twist the order-3 complex ω ($\mathbb{Z}_3$ generations on the quaternionic spinor) or the order-2 sign −1 ($\mathbb{Z}_2$ Möbius)? Pre-registered blind, then computed. RUBRICS done-bar below.

Blind pre-registration (committed to disk BEFORE any numpy)

Prediction: operative eigenvalue = ω, not −1. Reasoning: (1) three generations = rank+1 strata = odd cohomology h^1,h^3,h^5 (T1929), positions ∝ e^{2πik/3} → the generation-cycle is ORDER 3 → eigenvalues in {1,ω,ω^2}; −1 (order 2) cannot appear. (2) F872: the $\mathbb{Z}_2$ Möbius sign gives all three ODD generations the SAME sign → relative phase 0 → J=0; so −1 is the CP-KILLING assignment. (3) F874 Gate B: odd n_C → Sp(2) spinor pseudoreal (J^2=−1); a real reflection (±1, J^2=+1) is impossible on a quaternionic rep → the twist must be complex → the $\mathbb{Z}_3$ fixes it to ω, non-removable.

What I computed (two passes — the discipline caught the first)

Pass 1 (naive, REFUTED by the number): I stamped ω^k as a *global per-generation phase* on the spinor states. Result J ≈ 1e-43 ≈ 0. Correct catch: a per-generation global phase IS a field rephasing D_L, and the Jarlskog is rephasing-invariant BY CONSTRUCTION → it can never see a global row phase. The ω must enter the OVERLAP KERNEL (the complex positions feeding N(z,w)), not sit as a row phase. (Seventh subtlety on this map; several have been mine; running the number caught it before I claimed.)

Pass 2 (faithful type-IV Bergman kernel, F873 construction): generation k at complex point (ω^k · r_k)·u in $\mathbb{C}^5$; overlap V_ij = N(z_i,w_j)^{−6}/norm, N(z,w)=1−2(z,w̄)+[z,z][w̄,w̄]; up radii = saturation-inverted {1,α,α^2} (T2514), down = integer feed-down {1,3,5} (K1012); Löwdin→unitary; rephasing-invariant Jarlskog.

assignment	J	reading
ω position-twist (order 3, complex)	−0.0165 ≠ 0	CP present ⍰ (P1)
real positions (F498 control)	0 exactly	CP absent ⍰ (P2)
−1 position-twist (order 2, real (−1)^k)	0 exactly	−1 keeps positions REAL → J=0 ⍰ (F872)
ω under 6 random field rephasings	−0.0165 (invariant, dev 7e-18)	non-removable ⍰ (P3)

Structural half (operators): $\mathbb{Z}_3$ generation-cycle C eigenvalues = {1, ω, ω^2} — contains ω, does NOT contain −1; $\mathbb{Z}_2$ Möbius M = diag(−1,−1,−1). Quaternionic structure verified J^2=−1 on the Sp(2) spinor → no real (±1) reflection exists on this rep → the operative twist is forced complex.

The verdict (the pin)

Operative eigenvalue = ω (the $\mathbb{Z}_3$ generation phase on the Spin(5)=Sp(2) quaternionic spinor), NOT −1 (the $\mathbb{Z}_2$ Möbius sign). The −1 is real, keeps the positions real, and gives J=0 (that's F872, now re-derived as the CP-killing control). Only the order-3 complex ω enters the kernel as a genuine misalignment → J≠0, and the quaternionic $J^2=-1$ forbids rotating it back to a real ±1 (non-removable). This reconciles the whole thread: F872 (−1 → J=0), F873 (ω non-removable), F874 (odd n_C → quaternionic → complex twist forced), and now F875 (the operative eigenvalue is ω). CP existence is banked (det≠0, Elie); this identifies WHICH discrete object carries it.

RUBRICS Layer-2 done-bar (pin, honest)

- ⊠ Pre-registered the prediction BLIND to disk before computing (F875_blind_prereg.md).
- ⊠ Ran the number; Pass-1 naive model REFUTED (global phase = rephasing = J-invisible) — owned, fixed with the faithful kernel.
- ⊠ Pass-2 faithful kernel: P1 (ω→J≠0) ⍰, P2 (real & −1 → J=0) ⍰, P3 (non-removable) ⍰; structural half (order-3 carries ω not −1; quaternionic forbids real reflection) ⍰.
- ⊠ Tier: Structural-verified pin of the operative object/eigenvalue; CP EXISTENCE already banked (det≠0); magnitude stays OFF (J=−0.0165 here is model-dependent, NOT a prediction — the sign/value of δ_CP is a reverse-fit, K684).

Handoffs

- @Keeper — operative-spinor pin CONFIRMED blind: the banked CP phase is carried by the $\mathbb{Z}_3$ generation phase ω on the Spin(5)=Sp(2) quaternionic spinor, NOT the $\mathbb{Z}_2$ Möbius −1. −1 gives real positions → J=0 (that IS F872); only ω enters the kernel → J≠0, non-removable (quaternionic $J^2=-1$ forbids a real reflection). Ready to fold into the flagship (#31/#25) — this is the clean one-line statement of *why* CP is forced-complex. Magnitude off.
- @Elie — confirms your det≠0: the operative twist is ω (order-3), and I verified the literal −1 (order-2) is the J=0 branch, so we're not accidentally banking on the Möbius sign. When you wire the forced phase into CKM/PMNS (Lane 3), the phase enters through the COMPLEX POSITIONS in the FK overlap kernel, NOT as a per-generation global phase (that's a rephasing — it's invisible to J; I hit that and it cost me a pass). Coordinate the mixing-input build with me.
- @Cal — audit: I pre-registered blind, my first operationalization FAILED (global phase = rephasing), I owned it and fixed with the faithful kernel; the pass-2 result discriminates ω (J≠0) from real/−1 (J=0) cleanly and shows non-removability. Tier = Structural pin; magnitude OFF (J value is model-dependent, not a claim). No over-read.
- @Casey — the last CP question, settled, and I like how it settled. We asked: which "twist" actually carries the matter/antimatter asymmetry — the simple two-way sign-flip (like a Möbius strip's single half-turn, +1/−1), or the three-way turn that cycles the generations (a third of a full circle, ω)? I checked, blind: I wrote my answer down before computing. The two-way sign is REAL — it keeps the particle positions on the real number line, and a real configuration has exactly zero CP (that's our earlier proved result). Only the three-way turn is genuinely complex, and because our matter's spinor is the quaternion type (odd five → built-in $\sqrt{-1}$, no way to make it real), that three-way turn CANNOT be un-twisted back to a sign. So the asymmetry rides on the ω, the generation cycle — not the Möbius sign. One honest note: my first attempt got zero by mistake, because I put the twist in a place the physics literally can't see (a book-keeping phase you can always rotate away); running the number caught me, I moved the twist into the actual overlap, and it came out nonzero and un-rotatable, exactly as predicted. Eigenvalue ω, not −1. The value of the phase stays off the table — that's a reverse-fit. Verified at the source; nothing pushed.

Notes only; no toy/theorem claimed (blind pin verified). F875: operative-spinor pin. Blind pre-reg (committed before compute): operative eigenvalue = ω not −1. Structural: $\mathbb{Z}_3$ gen-cycle eigenvalues $\{1,\omega,\omega^2\}$ (has ω, not −1); Sp(2) spinor quaternionic $J^2=-1$ → no real reflection → twist forced complex. Kernel (faithful, F873): ω position-twist → J=−0.0165≠0 (CP); real positions → J=0 (F498); literal −1 (real (−1)^k) → J=0 (F872 CP-killer); ω non-removable under rephasings (dev 7e-18). CAUGHT: pass-1 naive global-phase model → J≈0 (global phase = rephasing = J-invisible), the 7th subtlety, fixed with kernel. VERDICT: banked CP carried by ω ($\mathbb{Z}_3$ generation phase on quaternionic spinor), NOT −1 ($\mathbb{Z}_2$ Möbius). Reconciles F872/F873/F874. Tier: Structural pin; magnitude OFF (J value model-dependent, reverse-fit). Nothing pushed. — Lyra